



«Approved»
by the Dean
Syzdykova Z.A.
«___» of _____ 2024

Syllabus
Academic Year 2024-2025

1. General Information	
Course title	Mathematics for Computational Science
Degree cycle (level)/major	Cyber Security
Year, term	1
Number of credits	5
Language of delivery:	English
Prerequisites	Numerical Methods
Postrequisites	Applied Mathematical Models
Lecturer(s)	Rysbaiuly Bolatbek, <u>DPh&MS, Professor</u> in Mathematical and Computer Modeling, rysbaiuly.b@gmail.com Astana IT University, Expo, C1 block, 3rd floor, office C1.1.332
2. Goals, Objectives and Learning Outcomes of the Course	
1. Course Description	Mathematics for Computational Science is a core course at Astana IT University; it plays an important role in understanding science, technology, economics, and computer science, among other disciplines. This course, Mathematics for Computational Science, covers difference derivatives, algorithms, and methods for solving the Cauchy problem for a first-order linear differential equation. It discusses various methods for solving a second-order linear differential equation with different boundary conditions. For the problems under consideration, it studies methods for solving various inverse problems. It investigates the convergence of iterative processes for solving the inverse problem. And it determines the convergence rate of the iterative processes.
2. Course Goal(s)	Mathematics for Computational Science is a core course at Astana IT University; it plays an important role in understanding science, technology, economics, and computer science, among other disciplines. This course, Mathematics for Computational Science, covers difference derivatives, algorithms, and methods for solving the Cauchy problem for a first-order linear differential equation. Methods for solving a second-order linear differential equation with various boundary conditions are considered. Convergence of approximate methods is numerically verified. Algorithms for solving the problem are created. For the problems under consideration, methods for solving various inverse problems are studied. Convergence of iterative processes for solving the inverse problem is investigated. And the convergence rate

	of the iterative processes is determined.
3. Course Objectives:	Postgraduate students should master the following concepts • Exact solution of linear differential equations of the first order. Solution of the inverse problem. Method, algorithm and program. • Exact solution of linear differential equations of the second order. Solution of the inverse problem. Boundary conditions: Dirichlet, Neumann and shifted boundary conditions. Method, algorithm and program. • • Approximate solution of nonlinear differential equations of the second order. Solution of the inverse problem. Boundary conditions: Dirichlet, Neumann and shifted boundary conditions. Method, algorithm and program
4. Skills & Competences	Upon completion of this course, graduate students should demonstrate competence in the following inverse problem solving skills: • Method of finding unknown parameters of processes described by first-order linear differential equations. Algorithm and program. • Method of finding unknown parameters of a process described by second-order linear differential equations under various boundary conditions. Algorithm and program. • Method of finding unknown parameters of a process described by second-order nonlinear differential equations. Algorithm and program.
5. Course Learning Outcomes:	By the end of this course, graduate students will be able to understand concepts related to inverse problems, namely: - finding unknown parameters of a process described by first-order linear differential equations. - finding unknown parameters of a process described by second-order linear differential equations. With Dirichlet, Neumann and smeared boundary conditions. - finding unknown parameters of a process described by second-order nonlinear differential equations.
6. Methods of Assessment	• Homework • Quiz • Midterm • Pre-Final and Final Exams
7. Reading List	Assigned reading materials and presentations should be read prior to class. Class lectures and discussions will proceed with supplemental and advanced topics, which could be difficult to understand unless students have read the assigned material. Readings are listed in the schedule section. All necessary updates and/or changes to the course will be reflected in the Learning Management System (moodle.astanait.edu.kz). <u>Basic Literature:</u> 1. Eric L. Miller. Fundamentals of Inverse Problems. Boston, 2003 2. Hanne Kekkonen and Yury Korolev Inverse Problems. Cambridge, 2019

	3. Kurt Bryan . An Intrjduction to Inberse Problems. 2011 4. Guillaume Bal Intrjduction to Inberse Problems.Chicago, 2019. <u>Supplementary literature:</u> 5. Мегралиев Я.Т. О разрешимости одной обратной краевой задачи для эллиптического уравнения второго порядка. Баку, 2011. 6. Су Линдэ, Численное решение обратных задач для параболических уравнений, 2019.
1. Resources	Online journals, articles, papers, books, and internet resources, as well as online emulators and online software for simulation.
2. Course policy	Course and University policies include: Attendance: Attendance is not allocated any grading points in the marking scheme, but is compulsory to pass the course. Normally, students are required to achieve course attendance of a minimum 70% to get admitted to the examination rubric. In case a student misses 30% or more class sessions without a valid excuse, the instructor has the right to mark him as “not graded.” In such case, a student is not admitted to the exam and automatically fails the course. It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more, he or she has to study material provided under the course on their own. Course instructors might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using the method of his or her choosing. Preparation for Class: Class participation is a very important part of the learning process in this course. Although it is not explicitly assessed, students will be assessed on the QUALITY of their contributions and ideas. Quality comments have one or more of the following properties: - Offer a different and unique, yet relevant, perspective; - Promote discussion and analysis; - Build on other comments. Classwork: Each lecture and practical lesson lasts 50 minutes for offline classes and 40 minutes for online classes. Students are expected to complete all readings and assignments ahead of time, attend class regularly, and participate in class discussions. In case of systemic student misconduct, the student can be dispensed from the classes. Being late on class: When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, lateness can become

chronic and spread throughout the class if left unchecked. Therefore, being late to class is not welcome and can be restriction activities by the course instructor.

Attestation I and II: Students who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course.

Homework / Assignments: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In the case of using someone's work (papers, articles, and other publications), all works must be properly cited. Failure to cite work will result in cheating from the students and may be the subject of additional disciplinary measures.

Late submissions: Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All grades are based on a percentage grading scale.

In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for submission. If approved, a new date will be given to the student depending upon the circumstances of the instructor.

Final exam

At the completion of this course, the Final Exam will be given in the written form

Laptops and mobile devices can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline, and the instructor will initiate appropriate action.

Online lessons can be used in case there won't be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run online lessons is Microsoft Teams for video calls and live webinars and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround.

Cheating and plagiarism are defined in the Academic conduct policies of the university and include:

1. Submitting work that is not your own papers, assignments, or exams;
2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation;
3. Submitting or using falsified data;

	4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work, and further actions will also be taken regarding the academic conduct policies of the university. Academic Conduct Policies of the university: The full texts of all the academic conduct codes will be posted to the students using the Learning Management System (moodle.astanait.edu.kz). Contacting the Course instructor: The easiest and most reliable way to get in touch with the course instructor is by email. Students must feel free to send an email if they have a question related to the course. The instructor responds as soon as they can but not always instantaneously. Besides that, students are also welcome to arrange a one-to-one meeting with the instructor during office hours to discuss the class offline and online.
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** Final exam can be given by the instructor in any other form such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

3. Course Content

#	Abbreviation	Meaning
1	ISIS	Instructor-supervised independent work
2	SIS	Students' independent work
3	IP	Individual project
4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Course topics	Lectures (h/w)	Practical sessions (h/w)	TSIS (h/w)	SIS (h/w)
1	Linear differential (LD) of the first order. Exact solution of the Cauchy problem. Mesh method. Difference derivatives of the first order. Right, left and central derivatives. Order of approximations. Checking the order of approximations. Algorithm	2	3	4	6
2	Application of first-order linear differential equation. Convergence. Algorithm and program. Additional condition. Inverse problem. Calculation of proportionality coefficient. Algorithm.	2	3	4	6

3	Linear DE of the second order $ay'' + by' + cy = f(t)$ Dirichlet boundary value problems $y(0) = A, y(T) = B$. Exact solution. Approximate solution. Inverse problem. $f(t)$-polynomial multiplied exponential function.	2	3	4	6
4	Linear differential equations of the second order $ay'' + by' + cy = f(t)$ Neumann boundary value problems $y'(0) = A, y'(T) = B$. Exact solution. Approximate solution. Inverse problem. $f(t)$-polynomial multiplied exponential function.	2	3	4	6
5	Linear differential equations of the second order $ay'' + by' + cy = f(t)$ Robin boundary value problems $y'(0) - ky(0) = A, y'(T) + hy(T) = B$ Exact solution. Approximate solution. Inverse problem. $f(t)$-polynomial multiplied exponential function.	0	3	4	6
5	Midterm	2	0	0	0
6	Linear differential equations of the second order $ay'' + by' + cy = f(t)$ Mixed boundary value problem $y(0) = A, y'(T) = B$. Exact solution. Approximate solution. Inverse problem. $f(t)$ is an exponential function multiplied by a trigonometric function.	2	3	4	6
7	Линейные ДУ второго порядка $ay'' + by' + cy = f(t)$ Смешанная краевая задачи $y'(0) = A, y(T) = B$. Точное решение. Приближенное решение. Обратная задача. $f(t)$-показательная функция умноженная на тригонометрическую функцию. Linear differential equations of the second order $ay'' + by' + cy = f(t)$ Mixed boundary value problem $y'(0) = A, y(T) = B$. Exact solution. Approximate solution. Inverse problem. $f(t)$ is an exponential function multiplied by a trigonometric function.	2	3	4	6

8	Linear differential equations of the second order $ay'' + by' + cy = f(t)$ Mixed boundary value problem $y'(0) = A, y'(T) + hy(T) = B$. Exact solution. Approximate solution. Inverse problem. $f(t)$ is an exponential function multiplied by a trigonometric function.	2	3	4	6
9	Linear differential equations of the second order $ay'' + by' + cy = f(t)$ Mixed boundary value problem $y'(0) - ky(0) = A, y(T) = B$. Exact solution. Approximate solution. Inverse problem. $f(t)$ is an exponential function multiplied by a trigonometric function.	0	3	4	6
9	Endterm	2	0	0	0
10	Nonlinear second-order differential equations $a(t)y'' + b(t)y' + c(t)y = f(t)$ Dirichlet boundary value problems $y(0) = A, y(T) = B$. Thomas method. Approximate solution. Inverse problem. $f(t)$ is an exponential function multiplied by a trigonometric function.	2	3	4	6
	Pre-final exam				
	Total hours: 150	20	30	40	60

3.2 List of Assignments for Student Independent Study

№	Assignments (topics) for independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	2	3	4	5
1	Linear differential (LD) of the first order. Exact solution of the Cauchy problem. Mesh method. Difference derivatives of the first order. Right, left and central derivatives. Order of approximations. Checking the order of approximations. Algorithm	10	Reading books, [3-5]	

2	Application of first-order linear differential equation. Convergence. Algorithm and program. Additional condition. Inverse problem. Calculation of proportionality coefficient. Algorithm.	10	Reading books, [3-5]	Exercises
3	Linear DE of the second order $ay'' + by' + cy = f(t)$ Dirichlet boundary value problems $y(0) = A, y(T) = B$. Exact solution. Approximate solution. Inverse problem. $f(t)$-polynomial multiplied exponential function.	10	Reading books, [3-5]	
4	Linear differential equations of the second order $ay'' + by' + cy = f(t)$ Neumann boundary value problems $y'(0) = A, y'(T) = B$. Exact solution. Approximate solution. Inverse problem. $f(t)$-polynomial multiplied exponential function.	10	Reading books, [3-5]	Exercises
5	Linear differential equations of the second order $ay'' + by' + cy = f(t)$ Robin boundary value problems $y'(0) - ky(0) = A, y'(T) + hy(T) = B$ Exact solution. Approximate solution. Inverse problem. $f(t)$-polynomial multiplied exponential function.	10	Reading books, [3-5]	Exercises
6	Linear differential equations of the second order $ay'' + by' + cy = f(t)$ Mixed boundary value problem $y(0) = A, y'(T) = B$. Exact solution. Approximate solution. Inverse problem. $f(t)$ is an exponential function multiplied by a trigonometric function.	10	Reading books, [3-5]	
7	Линейные ДУ второго порядка $ay'' + by' + cy = f(t)$ Смешанная краевая задачи $y'(0) = A, y(T) = B$. Точное решение. Приближенное решение. Обратная задача. $f(t)$-показательная	10	Reading books, [3-5]	Exercises

	функция умноженная тригонометрическая функция. Linear differential equations of the second order $ay'' + by' + cy = f(t)$ Mixed boundary value problem $y'(0) = A, y(T) = B..$ Exact solution. Approximate solution. Inverse problem. $f(t)$ is an exponential function multiplied by a trigonometric function.			
8	Linear differential equations of the second order $ay'' + by' + cy = f(t)$ Mixed boundary value problem $y'(0) = A, y'(T) + hy(T) = B$. Exact solution. Approximate solution. Inverse problem. $f(t)$ is an exponential function multiplied by a trigonometric function.	10	Reading books, [3-5]	Exercises
9	Linear differential equations of the second order $ay'' + by' + cy = f(t)$ Mixed boundary value problem $y'(0) - ky(0) = A, y(T) = B$. Exact solution. Approximate solution. Inverse problem. $f(t)$ is an exponential function multiplied by a trigonometric function.	10	Reading books, [3-5]	Exercises
10	Nonlinear second-order differential equations $a(t)y'' + b(t)y' + c(t)y = f(t)$ Dirichlet boundary value problems $y(0) = A, y(T) = B$. Thomas method. Approximate solution. Inverse problem. $f(t)$ is an exponential function multiplied by a trigonometric function.	10	Reading books, [3-5]	
Total hours		100		

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1 st attestation	Quiz	50	100
	Midterm	50	
2 nd attestation	Quiz	50	100
	Pre-Final	50	

Final exam	A written exam		100
Total	$0,3*1stAtt+0,3*2ndAtt+0,4*Final$		100

*** The number of assignments can be different. It depends from the course program and designed by course syllabus. But the total points of the assignment are 60 in each control period.*

** Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.*

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	
B	3,0	80-84	Good
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1,67	60-64	
D+	1,33	55-59	
D	1,0	50-54	Fail
FX	0	25-49	
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement

60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

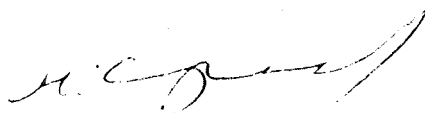
- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final assessment** is a combination of both individual (team) project (report) and oral presentation (slides) to evaluate the students' academic performance and professional skills. At the completion of this course each student has to submit an online project version conforming to the project outline, as well as to prepare and present a slide presentation that follows the presentation outline. Project iterations would be required to submit as well. Students should submit the written Report of Final project and slide-Presentation 3 days before the Final exam day.

If necessary, the teacher has the right to change points 3.1 and 3.2.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Rysbaiuly Bolatbek	Professor	02/09/2024	

Director



Sergaziyev M.Zh